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Name student:	Johan M. van Burik
Student number:	653303
Study Unit:	GP2
Lecturer(s):	Mulder, Rob; Diojdescu, Dan
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Section 1 Data Collection

Introduction

This section explains how the data for this study was collected before any analysis is conducted. Its purpose is to make the empirical foundation of the research explicit by clearly showing what data was gathered, from whom, from which project cases, and through which methods. The section presents the collected evidence first, while interpretation and analysis are carried out in Section 2.

The study was conducted within the Value Chain Hackers Lab at Windesheim University of Applied Sciences. The lab is a student led applied research initiative in which students, researchers, and external partners collaborate on real world supply chain challenges. Projects are carried out in an uncontrolled real world setting, where timelines, stakeholder access, and data availability vary across cases. This reflects practical research conditions rather than controlled experimentation.

At the time of this study, the lab consisted of 43 active students working across 12 teams, with 8 completed student projects forming the primary body of prior research activity. These projects produce structured research outputs such as reports, mappings, and repository based documentation. While no exact national count of comparable labs is established, the relevance of this context is supported through benchmarking against similar applied research environments and research data management practices identified in desk research.

The data collection follows the qualitative research design defined in the approved GP1. The original plan included desk research, case study analysis of completed projects, stakeholder interviews, and a short survey. During execution, one adjustment was made. The survey was not conducted due to timing and participation constraints and was replaced with additional structured interviews. This preserved the original intent of collecting multiple perspectives while improving depth and contextual quality of the data. The final dataset consists of 7 structured interviews, 5 documented case studies, and direct observation of the shared repository used to store and access project outputs. The interviews include 4 student project leads and 3 lab stakeholders in supervisory or coordination roles.

The documented case studies include the Bakery case (BBrood Supply Chain Network), which mapped the sustainable supply chain network of artisan bakeries in Amsterdam; the Cacao case (Cacao Chain Transparency Initiative), which explored cacao supply chains with a focus on sustainability and sourcing practices for smallholder farmers in Africa; the Textile case (Textile Twicely circular economy model), which examined a circular economy model for textile waste reduction and sustainable fashion supply chains; the CRM case (Critical Resource Management), which analysed supply chain structures and stakeholder systems for Gemeente Zwolle and local battery collection agencies; and the Organisational development case (VCH lab context), which explored the broader educational structure and culture within which projects are carried out. In addition, the Beer case (Circular Beer Packaging System), one of the completed projects, was analysed primarily through interview evidence due to limited accessible repository documentation. This results in coverage of 6 out of 8 completed projects.

The sampling size is explicitly justified in relation to the VCH Lab context. A typical project cycle involves approximately 10 to 15 students. The study includes 4 student project lead interviews, which represents between 26.7 percent of a 15 student cohort and 40 percent of a 10 student cohort. The focus on student project leads ensures that the data captures individuals directly responsible for structuring, executing, and documenting the research process. The case study sample covers 75 percent of all completed projects. Together, this provides sufficient depth and variation to identify recurring structural patterns in documentation, traceability, and reuse.

Desk research and field research are used in a complementary manner. Desk research identifies best practices in data traceability, reproducibility, and emerging developments such as AI supported knowledge management. Field research captures the current state within the Value Chain Hackers Lab through case studies, interviews, and repository observation. This ensures that the study is grounded in empirical evidence while also being benchmarked against external standards.

All data was collected using appropriate qualitative methods, including structured interviews, consistent case selection, and direct observation of project materials. Participants were informed about the purpose of the research, and all materials are documented and stored to ensure transparency and traceability. This ensures that the data collection process is methodologically sound, ethically conducted, and sufficient to support the analysis and conclusions presented in the following section.

To support transparency and traceability, all primary research materials are included in the appendices. Appendix A contains the full interview data, including transcripts and summaries of stakeholder input. Appendix B presents the detailed case study analyses and repository observations for each project. Appendix C documents the desk research and benchmarking sources that informed the study. These appendices function as the evidential foundation of the report, allowing the reader to trace how the collected data underpins the analysis and conclusions presented in Sections 2 and 3.

1.1 Research Context and Design Execution

The research design defined in GP1 was executed within the operational conditions of the Value Chain Hackers Lab, where projects develop in an uncontrolled real world environment. As a result, the research process did not follow a fixed sequence but adapted to the pace, structure, and constraints of ongoing student-led projects while remaining aligned with the original design.

The study was designed as a qualitative approach combining desk research, document analysis, stakeholder interviews, and a short survey. During execution, this structure was preserved in its core logic but adjusted for feasibility. Document analysis and interviews were conducted iteratively rather than in isolation, allowing early findings to guide further data collection and deepen understanding of documentation practices across cases.

Several practical constraints shaped execution. Access to project materials depended on what was stored in the shared repository, which varied across cases and often lacked intermediate outputs such as decision logs. This required a flexible approach, using available materials as entry points for further inquiry. In addition, stakeholder availability within the academic calendar influenced the timing and sequencing of interviews, requiring prioritisation of participants with central roles in project execution and oversight.

The evolving nature of the projects also affected execution. Documentation reflects decisions made under time pressure within fixed academic timelines rather than controlled research conditions. This required the study to treat documentation not only as outputs but also as evidence of how research processes unfolded in practice.

Despite these constraints, execution remained consistent with the original design logic. Multiple data sources were combined to capture different perspectives and enable cross checking of findings. This ensured methodological coherence while accurately reflecting real world conditions within the lab.

1.2 Data Sources and Collection Methods

This study uses a qualitative data collection strategy combining field research and desk research to ensure a complete and comparable dataset. This section outlines what data was collected and how it was obtained.

Field research establishes the current state of documentation, traceability, and reproducibility within the VCH Lab. Desk research provides the reference framework used to evaluate these practices.

Field research was conducted through three methods: case study document analysis, semi structured interviews, and repository observation.

Case study document analysis examined completed VCH Lab projects, including the Bakery, CRM, Cacao, Textile, and Organisational tracks. The data collected consists of preserved outputs such as reports, datasets, presentations, and available intermediate materials. This allows identification of which parts of the research process are documented and which are missing.

Semi structured interviews were conducted with former student project leads and key lab stakeholders. The interviews followed a structured protocol with predefined themes, while allowing respondents to explain their experiences in detail. The data collected includes explanations of how research was conducted, how decisions were made, and how documentation was handled during the projects.

Repository observation involved direct examination of the VCH Lab GitHub environment. The data collected includes folder structures, file naming, completeness of uploads, and the presence of intermediate files. This shows how research materials are stored and whether they can be accessed and understood.

These three methods form a triangulated dataset. Case analysis provides the documented outputs, interviews provide process explanations, and repository observation shows how materials are organised in practice.

Desk research provides the benchmark and conceptual reference for evaluation. It includes academic literature on reproducibility and research data management, institutional practices from universities and research infrastructures, and recent developments in AI supported research workflows.

Institutional benchmark data was collected from university based research infrastructures such as TU Delft, University of Twente, 4TU.ResearchData, and SURF Yoda, focusing on how documentation, metadata, and data management are structured in practice (TU Delft, 2023; University of Twente, 2022; 4TU.ResearchData, 2023; SURF, 2023).

Additional desk research examined the role of AI in research workflows, based on the GP2 Research Update Roadmap. This data shows how structured, well described data is required for retrieval, linking, and reuse in modern research systems (OpenAI, 2024; Turing Institute, 2023).

Together, these data sources provide a complete basis for analysing current practices and comparing them to established and emerging standard

1.3 Sampling, Ethics, and Data Quality

This study uses purposive sampling, selecting cases and participants based on their direct relevance to the research question. The objective is not statistical representation, but to generate in-depth, context-specific insight into documentation, traceability, and reuse within the VCH Lab (Patton, 2015).

Five completed projects were selected (Bakery, CRM, Cacao, Textile, and Organisational), as they represent full research trajectories with accessible outputs. Analysing multiple cases enables the identification of recurring patterns across projects, strengthening analytical validity (Yin, 2018), while variation across domains reduces the risk of context-specific bias.

The study includes seven interviews: four student project leads and three lab stakeholders—Maxime Bouillon (Teacher/Researcher), Rea Vaz (Researcher/Lab Coordinator), and Christiaan Verhoef (Technical Facilitator/Lab Coordinator). Project leads were selected due to their direct responsibility for structuring research and documentation, while stakeholders provide system-level perspectives on supervision, coordination, and reuse. Together, these roles capture both the production and evaluation of documentation practices (Nonaka & Takeuchi, 1995; Kralj & Landwehr, 2025).

Sample adequacy is justified relative to the lab context. A typical VCH Lab cycle includes 10–15 students, meaning the seven interviews represent approximately 47–70% of a cohort while covering the most central roles. Combined with five case studies, this provides sufficient depth and variation to identify structural patterns within the documentation lifecycle (Patton, 2015; Yin, 2018).

Data quality is ensured through triangulation of case study analysis, interviews, and repository observation. Project materials provide the empirical record of what is preserved, interviews explain how and why documentation practices occur, and repository observation shows how materials function in practice. The consistency of findings across these sources—such as missing intermediate reasoning, limited decision traceability, and fragmented storage—supports the validity of the dataset. Reliability is further strengthened through iterative comparison during data collection, where emerging insights were continuously cross-checked across sources.

The credibility of the data is reinforced by the use of authentic project outputs and participants directly involved in project execution or coordination. Supporting desk research draws on established literature and institutional frameworks in reproducibility and research data management (Kallio et al., 2016; DeJonckheere & Vaughn, 2019; Wilkinson et al., 2016; OECD, 2021), ensuring alignment with recognised standards.

Ethical considerations were addressed throughout. Participation was voluntary, participants were informed of the research purpose, and no sensitive personal data was collected. The analysis focuses on structural processes rather than individual performance, and all primary materials are traceable in the appendices to ensure transparency and auditability.

Limitations are acknowledged. Project repositories often lacked intermediate documentation, particularly regarding decision-making and analytical steps; however, these absences form part of the empirical evidence, as incomplete documentation is central to the problem investigated (Feger et al., 2019; Lalu et al., 2023). In addition, interviews rely partly on retrospective accounts, which may introduce recall bias. This was mitigated through systematic comparison with project materials and repository structures.

Overall, the dataset is sufficient, relevant, and methodologically robust. The combination of purposive sampling, triangulated data sources, and iterative validation provides a reliable foundation for the analysis that follows (Patton, 2015; Yin, 2018).

1.4 Data Collection Execution

This section explains **how the data was collected in practice**, ensuring that the process is transparent, traceable, and reproducible. It specifies how interviews, case studies, and repository observations were conducted, what materials were examined, and how the data was recorded and stored.

Interviews

The study conducted **seven interviews** using a **structured format with controlled flexibility**. The interview guide was developed based on prior desk research on research data management, lifecycle processes, and documentation practices, ensuring that all questions addressed relevant constructs. Selected questions allowed participants to elaborate freely, enabling the capture of context-specific insights while maintaining consistency across interviews (Kallio et al., 2016; DeJonckheere & Vaughn, 2019). Interviews were conducted with **four former student project leads** and **three lab stakeholders (Maxime Bouillon, Rea Vaz, and Christiaan Verhoef)**. Each interview focused on documentation practices during project execution, decision-making processes, and how outputs were later accessed or reused. Interviews were recorded and transcribed, and full transcripts are included in **Appendix A**, ensuring traceability of all interview evidence.

Case Study Document Analysis

Case study analysis was conducted on **five completed VCH Lab projects**: Bakery, CRM, Cacao, Textile, and Organisational development. The analysis examined **existing project documentation** stored in the repository, including reports, presentations, visual outputs, and supporting files. Each case was reviewed to determine what outputs were preserved, how documentation was structured, which parts of the research process were visible, and which elements were missing. This enables direct assessment of documentation practices based on observable materials (Yin, 2018). The full case analysis is provided in **Appendix B**.

Repository Observation

In addition to case-level analysis, **direct observation of the shared repository structure** was conducted. This involved examining how materials were organised, named, and accessed across projects. The focus was on whether stored documentation could be understood and reused without prior project involvement, providing insight into how documentation functions in practice after project completion.

Data Collection Context

Data collection was conducted during a **VCH Lab project cycle**, with interviews carried out between **January and early February**. This ensured that both project materials and participant reflections were situated within the same operational context of the lab.

Data Storage and Traceability

All collected data is systematically documented to ensure **traceability and auditability**. Interview transcripts are included in **Appendix A**, case study analysis in **Appendix B**, desk research mapping in **Appendix C**, and methodological justification in **Appendix D**. All original project materials remain in the VCH Lab repository.

1.5 Observations from Project Documentation and Interviews

This section presents the observed condition of project documentation and interview evidence across the VCH Lab prior to analysis. It outlines what information is available, how it is structured, and what is missing, based on project materials and participant accounts, establishing the empirical foundation for the analysis that follows.

Observations from Project Documentation

Across the five documented cases (Bakery, CRM, Cacao, Textile, and Organisational development), final project outputs are consistently present and accessible. These outputs, including reports, visual mappings, and presentations, indicate that substantial analytical work was conducted in each project.

In contrast, intermediate stages of the research process are not systematically preserved. Data collection materials such as interview records or datasets are inconsistently stored or absent, while analytical processes, including how information was interpreted, compared, or synthesised, are rarely documented. Changes in project direction and key decisions are also not explicitly recorded, making it difficult to reconstruct how the research evolved.

As a result, there is a clear distinction between what was produced and how it was produced. Final outputs are visible, but the process leading to them is largely not reconstructable. Where supporting materials exist, their role within the analysis is often unclear, limiting their usefulness. Documentation structures also differ significantly between projects. There is no consistent approach to file organisation, naming, or linking between materials. Some projects contain loosely organised folders with mixed content, while others include only final deliverables. This lack of consistency reduces comparability and makes navigation difficult without prior involvement.

Observations from Interviews

Interview data explains how these documentation conditions emerge in practice. Across all seven interviews, documentation is consistently described as secondary to producing final deliverables rather than as an integrated part of the research process. Student project leads report that intermediate steps such as analytical reasoning, iterations, and discarded approaches are rarely documented. Changes in project direction are also not consistently recorded. As a result, much of the research process remains tacit and embedded in individual understanding.

Lab stakeholders confirm these limitations from a reuse perspective. They report that project outputs are difficult to interpret independently because the reasoning behind decisions is not visible. In practice, understanding prior work often depends on direct communication with the

original team, which limits reuse and continuity. These findings indicate that the observed documentation gaps are structural rather than incidental.

Repository and Use in Practice

Observation of the shared repository reinforces these findings. While materials are centrally stored, their organisation does not support interpretation or reuse. Files often lack consistent structure, naming, and contextual information, making it difficult to determine their relevance or relationship to the overall project. As a result, materials cannot be reliably understood without prior knowledge. The repository therefore functions primarily as a storage system for outputs rather than as a structured representation of the research process.

Cross-Case Empirical Patterns

Across all cases, consistent patterns emerge. Final outputs are preserved, but intermediate processes are not. Decision making and analytical reasoning are rarely documented. Traceability between sources, analysis, and conclusions is limited. Documentation structures are inconsistent, and understanding depends on tacit knowledge rather than accessible records. The repository supports storage but not systematic retrieval or interpretation. These patterns are confirmed across cases and data sources, indicating that they reflect structural characteristics of current practice rather than isolated issues.

1.6 Data Quality and Limitations

The dataset was sufficient to identify consistent patterns, with certain limitations present. Project repositories were incomplete, with missing intermediate steps and limited traceability, meaning the full research process could not be reconstructed from documentation alone. The planned survey was replaced with structured interviews due to timing constraints, improving depth but reducing breadth. Potential bias exists in retrospective interview data from project leads. This was mitigated through triangulation, in which findings were cross-checked against case documentation, interviews, and repository observations. Despite these limitations, the use of multiple data sources provides a consistent and reliable basis for analysis.

1.7 Link to Research Question

The collected dataset directly supports answering the main research question: how the VCH Lab can design and validate a reproducible research framework that ensures transparency, verifiability, and reuse of student research. The field data provides an empirical view of the current research practice (IS situation). Case study analysis and repository observation reveal how documentation is structured, where traceability breaks down, and how research processes are inconsistently executed. The interviews complement this by explaining how decisions are made in practice and why key elements are not consistently recorded. This enables the identification of structural barriers to reproducibility, including fragmented documentation, weak

linkage between data and conclusions, and limited continuity across project cycles. These findings directly address the diagnostic component of the research process. In parallel, desk research and institutional benchmarking define the required conditions for reproducible research, including structured documentation, lifecycle integration, and traceable decision-making practices. These benchmarks establish the target state (SOLL situation) and provide validated criteria for framework design (OECD, 2021; Science Europe, 2021).

Together, the dataset enables a structured comparison between the current and desired state. This forms the basis for the gap analysis in Section 2, where identified deficiencies are translated into concrete design requirements and validation criteria for the reproducible research framework.

2.1 Analytical Approach and Reasoning Logic

This section defines how the collected data was analysed in order to identify patterns, establish the current and ideal states of data traceability, and derive the gaps that inform system requirements. The purpose is to make the analytical logic explicit while maintaining a clear and efficient structure.

The analysis follows a **comparative approach** between two states. The ideal state is derived from desk research (Section 2.2), based on established research data management practices, reproducibility frameworks, and institutional benchmarks. The current state is derived from field research within the Value Chain Hackers Lab (Section 2.3), using case study documentation, interviews, and repository observation. Comparing these two states enables the identification of structural differences in how data traceability is achieved in practice.

To ensure consistency, both states are analysed using the same six dimensions: **data creation, storage structure, decision traceability, process visibility, retrieval and use, and continuity**. These dimensions represent the full research lifecycle and provide a shared analytical framework that allows direct comparison between current practices and established standards.

The field research combines three complementary data sources. Project repositories show what materials were preserved, interviews explain how documentation decisions were made, and repository observation shows how materials can be accessed and interpreted by users without prior involvement. Analysing these sources together ensures that the findings reflect both documented outputs and the underlying research process.

Findings are based on **pattern validation across cases and data sources**. A pattern is only considered relevant when it appears across multiple projects or is supported by both documentary and interview evidence. This ensures that conclusions reflect structural conditions within the lab rather than isolated cases.

The analysis proceeds in three steps. First, patterns are identified within the field data. Second, these patterns are compared with the ideal state defined through desk research. Third, the differences between the two are interpreted as structural gaps in the data traceability process.

This analytical approach ensures that the transition from empirical evidence to gap identification and requirement definition remains consistent, explicit, and grounded. It provides the foundation for Section 2.3, which defines the current state, and Section 2.4, which translates the identified gaps into system requirements.

2.2 Desk Research Analysis: Defining the Ideal State

This section defines the **ideal state of data traceability** by analysing established practices from academic literature, institutional frameworks, and research infrastructures across the Netherlands, Europe, and international contexts. The purpose is to derive a clear and operational benchmark that allows direct comparison with current practices in the Value Chain Hackers Lab (VCH).

The desk research draws on reproducibility research, research data management (RDM) standards, and governance frameworks, including guidelines from organisations such as OECD, Science Europe, and national frameworks such as NWO. In addition, infrastructure practices from platforms such as SURF and 4TU.ResearchData are considered, as well as principles such as the FAIR principles. These sources represent widely adopted standards for ensuring that research remains traceable, reproducible, and reusable across institutional contexts.

Rather than treating these sources separately, the analysis identifies **recurring structural requirements** across them. This ensures that the ideal state is not theoretical, but grounded in how data traceability is operationalised in comparable research environments. These recurring requirements are synthesised into six analytical dimensions that define the full data traceability process and provide a consistent basis for comparison with the VCH context.

Data creation

Research data must be captured together with its context and interpretation at the moment of creation, including raw data, intermediate analysis, and outputs. This ensures that analytical meaning is preserved and conclusions remain verifiable (Peng, 2011; Wilkinson et al., 2016).

Storage structure

Data must be organised within a structured and lifecycle-based system, supported by consistent naming, metadata, and clear relationships between files. Institutional infrastructures such as SURF and 4TU operationalise this through predefined repository structures and metadata requirements (OECD, 2021; Science Europe, 2022).

Decision traceability

Key research decisions must be explicitly documented and linked to supporting evidence. This ensures that conclusions can be understood, evaluated, and reproduced, aligning with reproducibility standards and FAIR requirements (Wilkinson et al., 2016).

Process visibility

The full research process—including intermediate steps, iterations, and validation stages—must remain visible and reconstructable. Reproducibility frameworks emphasise that knowledge is created through these intermediate stages and cannot be recovered from final outputs alone (Peng, 2011; Nosek et al., 2015).

Retrieval and use

Data must be findable, accessible, and interpretable without prior involvement. This depends on structured organisation, metadata, and clear contextualisation. Increasingly, this also supports machine-assisted retrieval and analysis, as AI systems rely on well-structured and well-described data (European Commission, 2024; UNESCO, 2023).

Continuity

Research outputs must be preserved in a way that supports reuse and extension across projects. Institutional and governance frameworks emphasise that long-term value is created when knowledge accumulates rather than being recreated (OECD, 2021; Science Europe, 2022).

Together, these dimensions define a coherent ideal state in which data, context, decisions, and processes are systematically captured, structured, and preserved. In such a system, a user—human or computational—can independently locate materials, understand how research was conducted, trace how conclusions were derived, and reuse outputs without reliance on the original researchers.

This ideal state reflects how data traceability is implemented in leading research environments and infrastructures across national and international contexts. It therefore provides a **validated benchmark** against which the current practices within VCH can be analysed. By applying the same dimensions in Section 2.3, the study enables a direct comparison between what occurs in practice and what is required to achieve traceable, reproducible, and reusable research.

2.3 Field Research Analysis: Defining the Current State

Purpose and Analytical Approach

This section defines the current state of data traceability within the Value Chain Hackers Lab (VCH) based on field research. Using the analytical framework established in Sections 2.1 and 2.2, it examines how research data is created, structured, documented, and transferred in practice. The analysis draws on three integrated sources: project repositories (GitHub), interviews with students and supervisors, and direct observation of how materials can be accessed and interpreted after project completion. In addition, the analysis is supported by the

VCH Lab Case Study Insight Analysis (Appendix B), which provides a consolidated overview of case-level evidence and structural documentation patterns across projects.

These sources are analysed together to reconstruct how the research process unfolds in practice, including both what is visible and what is structurally missing. Findings are based on patterns validated across multiple cases and data sources, ensuring that they reflect systemic conditions rather than isolated project outcomes.

Case Evidence

(All outputs referenced were retrieved from the VCH Lab GitHub repository and analysed in combination with interview data and observational review.)

Project	Available Outputs (GitHub Repository)	Missing Documentation	Implication for Traceability
Bakery	• Audio stakeholder interview • Final presentation slides	• Interview transcript • Desk research & preparation notes • Analytical steps and interpretation • Internal team notes	Data and outputs are present, but the analytical transformation is not documented, resulting in a breakdown between evidence and conclusions
CRM	• Presentations & slide decks • System maps & conceptual models • Draft documents and working files	• Interview transcripts • Structured analysis documentation • Version history • Source-to-claim mapping • Centralised repository structure	High output volume without structured linkage between stages, making the research process non-reconstructable
Cacao	• Final written guide • Collection of sources • Draft versions	• Notes comparing sources • Documentation of source selection • Evidence-to-claim mapping • Analytical reasoning steps	Outputs are coherent, but the absence of documented synthesis limits verification and reuse
Textile	• Visual boards (Miro, Canva) • Presentations and mapped insights	• Exported/archived outputs • Source references • Written analysis • Documentation of reasoning	Research remains tool-dependent and lacks independent documentation, preventing transferability and traceability

Organizational

- Master presentation deck
- Photos of final presentation

- Research documentation
- Project files or datasets
- Analytical outputs
- Defined documentation standards

Absence of system-level structure results in inconsistent and non-standardised documentation practices across the research lab

Cross-Case Patterns

Across all cases, consistent structural breakdowns emerge across the data traceability process. Documentation is not embedded within the research process but concentrated at the output stage, meaning intermediate steps such as preparation, analysis, and iteration are not preserved, creating a disconnect between raw data and final outputs. Decision-making remains largely tacit, as reasoning occurs during projects but is not systematically recorded, making key decisions non-traceable beyond the original team. Storage does not reflect the research lifecycle; materials exist but are fragmented across tools and formats, without a structure linking data, analysis, and outputs into a coherent sequence. As a result, provenance chains are broken, and it is not possible to trace how specific inputs lead to specific outputs because analytical transformations are not documented. Retrieval depends on prior knowledge rather than documentation, as materials lack sufficient context for independent interpretation, limiting usability and verification. Continuity is also not achieved; because reasoning, decisions, and context are not preserved, projects cannot be reliably reused or extended, preventing cumulative knowledge development. Taken together, these patterns show that breakdowns occur systematically across all stages of the research lifecycle rather than at isolated points.

Synthesis of the Current State

The current state of VCH can be defined as an output-oriented research environment in which process traceability is not systematically preserved. Research activities clearly occur and produce meaningful outputs, but the connections between data, reasoning, and conclusions are not maintained: data lacks context, storage is fragmented, decisions are not explicitly recorded, and the research process is not fully visible. As a result, outputs can be accessed but cannot be independently understood, verified, or reused. This indicates that the limitation is not a lack of research activity, but a lack of structural conditions that support traceability across the full research lifecycle. Documentation is not embedded within the workflow, resulting in tacit decision-making, broken provenance chains, and limited reuse. The VCH Lab therefore operates as a series of isolated project outputs rather than a continuous and reproducible research system. This definition of the current state provides the analytical basis for Section 2.4, where these structural limitations are compared to the ideal state to identify gaps and derive system requirements.

2.4 Gap Analysis and Derivation of Data Traceability Requirements

Purpose and Analytical Approach

This section translates the findings from Section 2.3 into a structured gap analysis by comparing the current data traceability process within the Value Chain Hackers Lab (VCH) with the ideal state defined in Section 2.2. The aim is to identify what is missing, why these gaps limit traceability, reproducibility, and reuse, and what structural conditions are required to address them.

The analysis applies six dimensions—data creation, storage structure, decision traceability, process visibility, retrieval and use, and continuity—to ensure a consistent comparison between field evidence and institutional benchmarks. The current state is derived from case documentation, interviews, and repository observation, while the ideal state is grounded in reproducibility literature, research data management standards, and institutional practices.

Gap Analysis: Current vs Ideal

Dimension	Current State (VCH)	Ideal State (Benchmarks)	Gap	Implication
Data Creation	Data generated without consistent context or interpretation	Data captured with context and analytical meaning	Data–meaning separation	Limits interpretability and validation (Wilkinson et al., 2016; Patton, 2015)
Storage Structure	GitHub storage present but fragmented and inconsistent	Structured, lifecycle-based storage with clear relationships	No lifecycle structure	Reduces findability and usability (Mons et al., 2017)
Decision Traceability	Decisions implicit and undocumented	Decisions explicitly recorded and linked to evidence	Missing decision traceability	Prevents verification of conclusions (Peng, 2011)

Process Visibility	Intermediate steps not consistently documented	Full process visible and reconstructable	Limited process visibility	Limits reproducibility and learning (Nosek et al., 2015)
Retrieval & Use	Outputs accessible but not independently interpretable	Outputs understandable without prior involvement	Limited independent usability	Reduces reuse potential (Wilkinson et al., 2016)
Continuity	Knowledge not reliably transferred across projects	Knowledge preserved for reuse and extension	Lack of embedded continuity	Prevents cumulative knowledge (OECD, 2021)

Synthesis and Derivation of System Requirements

The analysis shows that while the Value Chain Hackers Lab (VCH) consistently produces meaningful research outputs, it lacks a structured system that connects data, analysis, and conclusions. Three key patterns define the current state. **First**, breakdowns occur between stages: while data is created and outputs are produced, the links between data, reasoning, and outcomes are not preserved. **Second**, storage supports access but not understanding: materials are available in the repository, but lack the structure and context required for independent interpretation. **Third**, knowledge remains individualized rather than routinely and systematically inherent, meaning that reuse depends on the original researchers rather than the documentation itself.

These patterns indicate that the limitation is structural rather than incidental. The issue is not the absence of research activity, but the absence of a system that ensures traceability across the full research lifecycle. The identified gaps are therefore interconnected. When data is not recorded with context, storage becomes fragmented. Fragmented storage obscures decisions, which reduces process visibility. Without process visibility, outputs cannot be independently interpreted, and continuity cannot be achieved. This chain explains how individual breakdowns collectively prevent reproducibility and reuse.

To address this, documentation must move from fragmentation to an integrated system that ensures information is reliably documented and accessible. Based on this analysis, six core requirements have been identified as essential to enable traceable, reproducible, and reusable research within VCH:

1. Structured Data Capture

Data must be recorded together with its context and interpretation at the point of creation.

2. **Lifecycle-Based Storage**

Materials must be organised according to the progression of the research process within a shared system.

3. **Explicit Decision Logging**

Key decisions must be documented and directly linked to supporting evidence.

4. **Full Process Visibility**

The research process must be documented as a continuous and reconstructable sequence.

5. **Contextualised Outputs**

Outputs must include sufficient context to be independently understood and reused.

6. **Embedded Continuity**

Research must be preserved in a structured system that enables reuse and extension across projects.

Together, these requirements define the structural conditions needed to establish an effective data traceability process. They translate the identified gaps into actionable design principles and provide a direct foundation for the system proposed in Section 3.

3.1 Introduction to the Conclusions

This section translates the results of the data analysis into a set of conclusions and design criteria. Its purpose is not to propose a solution, but to define what a solution must achieve to address the structural issues identified in the Value Chain Hackers Lab.

Section 2 showed that current research practices do not ensure transparency, verifiability, or reuse. While final outputs are available, the underlying research process is not systematically documented, making reconstruction and reuse difficult. The analysis confirms that this is a structural issue. Documentation is not embedded throughout the research process, whereas desk research shows that effective research environments integrate structured and continuous documentation. The gap between these states defines the problem addressed in this project.

Based on this gap, core aspects were identified in Section 2. This section translates those aspects into explicit and measurable criteria. The conclusions remain at the level of specifications, defining what must be achieved without prescribing how. These criteria form the foundation for the next phase of solution development.

3.2 From Analysis to Core Conclusions

This section presents the core conclusions derived from the data analysis conducted in Section 2. These conclusions are based on the comparison between the current state observed in the Value Chain Hackers Lab and the ideal state identified through desk research and benchmarking. Each conclusion directly contributes to answering the main research question:

“How can a structured and reproducible research framework be designed to improve data traceability, transparency, and reuse within student-led supply chain research at the Value Chain Hackers Lab?”

The conclusions are grounded in field research, including the VCH Lab case study analysis (Appendix B) and interviews with student project leads and lab stakeholders (Appendix A), and are supported by desk research on reproducible research practices and data management standards (Appendix C).

Core Conclusions Overview

Conclusion	Key Observation	Implication
Research processes are not structurally externalised	Final outputs are available, but intermediate steps such as decision making, data transformations, and analytical reasoning are not documented	Research cannot be reconstructed or verified, limiting transparency, validation, and methodological learning
Data storage is fragmented and lacks consistent structure	Project data is distributed across multiple tools and locations without consistent organisation or naming conventions	Users cannot reliably locate, interpret, or connect project materials, reducing usability and access
There is no standardised approach to metadata and indexing	Files lack descriptions, contextual information, and links to related components within the project	Research outputs remain context dependent and difficult to interpret or reuse
The research lifecycle is not embedded in practice	Documentation occurs inconsistently and is not integrated throughout the research process	Gaps emerge between data collection, analysis, and outputs, breaking traceability across the lifecycle
Research outputs are not designed for reuse or continuity	Completed projects lack sufficient structure and contextual information for future use	Each new project starts from scratch, limiting cumulative

knowledge building and long term value		
There is a lack of socio technical alignment supporting research practices	There are no enforced standards, integrated tools, or evaluation criteria that require documentation and traceability	Documentation practices remain inconsistent and dependent on individual behaviour rather than a shared system

3.3 From Conclusions to Design Requirements

This section translates the core conclusions identified in Section 3.2 into concrete design requirements for the proposed research framework. Each requirement is directly derived from the identified structural gaps and defines what must be implemented to achieve a reproducible, traceable, and reusable research process within the Value Chain Hackers Lab.

The purpose of this section is to ensure that the design of the solution is not based on assumptions, but is fully grounded in empirical findings and aligned with the ideal state identified through desk research.

Design Requirements Derived from Core Conclusions

Conclusion	Design Requirement	What Must Be Implemented
Research processes are not structurally externalised	The research process must be fully documented and externalised throughout the project lifecycle	A structured system that captures decision making, data transformations, and analytical reasoning at each stage of the research process
Data storage is fragmented and lacks consistent structure	All research data must be stored within a unified and standardised structure	A centralised storage environment with predefined folder structures, naming conventions, and consistent organisation across all projects

There is no standardised approach to metadata and indexing	All research outputs must be supported by structured metadata and indexing	A metadata framework that describes content, context, relationships, and origin of each file or dataset
The research lifecycle is not embedded in practice	Documentation and traceability must be integrated into each stage of the research lifecycle	A workflow in which documentation is embedded from data collection through analysis to final output, rather than added retrospectively
Research outputs are not designed for reuse or continuity	Research outputs must be structured to support reuse and continuity across projects	A system that ensures outputs are complete, interpretable, and accessible for future users without prior project involvement
There is a lack of socio technical alignment supporting research practices	The research environment must enforce and support standardised documentation practices	Integration of tools, guidelines, and evaluation criteria that require and enable consistent documentation behaviour

Synthesis of Design Requirements

The requirements defined above demonstrate that improving data traceability within the Value Chain Hackers Lab is not limited to improving storage practices alone. Instead, it requires the implementation of a structured research system in which documentation, organisation, and traceability are embedded throughout the entire research process.

The findings show that each identified issue is interconnected. Fragmented storage, missing metadata, and lack of process visibility all stem from the absence of a system that integrates people, processes, and tools. As a result, the design requirements must be implemented as part of a coherent framework rather than as isolated improvements.

These requirements establish the foundation for the design criteria defined in the next section, where each requirement is translated into measurable and testable specifications for the proposed solution.

3.4 Design Criteria and Specifications

This section translates the design requirements defined in Section 3.3 into measurable and testable design criteria. These criteria specify how the proposed research framework must perform in practice to ensure improved data traceability, transparency, and reuse within the Value Chain Hackers Lab.

Each criterion defines what successful implementation looks like and establishes clear conditions under which the framework can be evaluated. The criteria are grounded in the empirical findings from Section 3.2, aligned with the design requirements from Section 3.3, and supported by established principles from desk research, including reproducible research practices, metadata standards, and lifecycle-based data management.

Design Criteria and Specifications

Design Requirement	Design Criterion	Specification (Measurable Outcome)
The research process must be fully documented and externalised	Research steps are consistently documented and traceable	All projects include documented records of decision making, data transformations, and analytical steps that allow full reconstruction of the research process
All research data must be stored within a unified and standardised structure	Data storage is consistent and navigable across projects	All project files follow a predefined folder structure and naming convention, allowing users to locate key materials within a fixed and predictable structure
All research outputs must be supported by structured metadata and indexing	Research materials are clearly described and contextually linked	Each file or dataset includes metadata describing its content, origin, purpose, and relationship to other project components
Documentation and traceability must be integrated into each stage of the research lifecycle	Traceability is maintained throughout the research process	Documentation is present at each stage of the project, from data collection to final output, without gaps between stages
Research outputs must be structured to support reuse and continuity	Outputs are reusable without prior project involvement	External users can understand and reuse project materials without needing additional explanation from original project members

The research environment must enforce and support standardised documentation practices

Documentation practices are consistently applied across projects

All projects follow the same documentation standards, supported by tools, templates, and evaluation criteria that ensure compliance

Synthesis of Design Criteria

The design criteria demonstrate that the effectiveness of the proposed framework depends on its ability to operationalise traceability as a consistent and measurable aspect of the research process. Rather than treating documentation as an optional activity, the criteria require it to be embedded, standardised, and verifiable across all projects. Each criterion defines a clear condition that must be met for the framework to function as intended. Together, they ensure that research processes become visible, data becomes structured and interpretable, and outputs become reusable beyond the scope of individual projects.

By translating abstract requirements into measurable specifications, this section establishes a direct link between the identified problems and the evaluation of the proposed solution. This ensures that the framework can be tested, validated, and refined in practice, rather than remaining a conceptual model.

3.5 Prioritisation of Criteria

This section defines the priority and implementation sequence of the design criteria presented in Section 3.4. The prioritisation is based on two principles. First, the dependency between criteria, where certain conditions must be established before others can function. Second, the chronological order in which the framework can be realistically implemented within the Value Chain Hackers Lab.

The criteria are therefore not treated as isolated elements, but as an interconnected system in which each layer enables the next.

Prioritised and Sequenced Design Criteria

Phase

Priority
Level

Design Criterion

Dependency and Rationale

Phase 1: Structural Foundation	High	Data storage is consistent and navigable across projects	A unified structure is the starting point of the system. Without a fixed organisational logic, documentation and metadata cannot be applied consistently
Phase 1: Structural Foundation	High	Research steps are consistently documented and traceable	Once structure exists, research processes must be captured within it. This establishes visibility of how outputs are produced
Phase 2: Lifecycle Integration	High	Traceability is maintained throughout the research process	After documentation is introduced, it must be embedded across all stages of the research lifecycle to prevent gaps between phases
Phase 3: Contextualisation	Medium	Research materials are clearly described and contextually linked	Metadata and indexing add meaning to structured and documented data, enabling interpretation and navigation
Phase 3: Contextualisation	Medium	Documentation practices are consistently applied across projects	Standardisation ensures that all projects follow the same approach, reinforcing consistency and reducing variability
Phase 4: System Outcome	Low	Outputs are reusable without prior project involvement	Reusability is the result of all previous criteria being fulfilled. It cannot be achieved without structure, documentation, lifecycle integration, and metadata

The prioritisation demonstrates that the design criteria form a sequential and interdependent system rather than a set of independent improvements. The first phase establishes the structural foundation, where data is organised and research processes become visible. Without this foundation, higher level functions such as metadata, standardisation, and reuse cannot operate effectively.

The second phase ensures that traceability is not limited to isolated steps but is embedded throughout the research lifecycle. This prevents fragmentation between stages and creates continuity across the research process.

The third phase introduces contextualisation and consistency. Metadata provides meaning to the documented data, while standardisation ensures that this approach is applied uniformly

across projects. Together, these elements enable the system to function reliably across different users and cohorts.

The final phase represents the outcome of the system. Reusability emerges when all preceding conditions are met, allowing research outputs to be understood, accessed, and built upon without requiring prior involvement in the original project.

This prioritised and chronological structure ensures that the proposed framework can be implemented in a realistic and scalable way, where each phase builds directly on the previous one and contributes to resolving the structural issues identified in Section 3.2.

3.6 Conclusion

This section translated the findings from the data analysis into a structured set of design requirements and measurable criteria for improving data traceability within the Value Chain Hackers Lab. By systematically moving from identified gaps to concrete specifications, the section established a clear and evidence based foundation for the proposed research framework. The conclusions demonstrated that the current research environment lacks the structural conditions required for reproducibility, transparency, and reuse. In response, the design requirements defined what must be implemented to address these issues, while the design criteria specified how successful implementation can be evaluated in practice. The prioritisation further clarified the order and interdependence of these elements, ensuring that the framework can be implemented in a logical and feasible manner. Together, these components form a coherent design logic in which each step builds on the previous one, from problem identification to measurable solution conditions. This ensures that the proposed framework is not only conceptually grounded, but also practically applicable and testable within the research environment. The next section builds on this foundation by presenting the final framework design, in which the defined criteria are operationalised into a structured and implementable research workflow.

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